Natural Language - Project report

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1 Introduction

We assign clinical records to one of 12 medical specialties, with 2,662 training and 398 test records adapted from a Kaggle dataset[1]. Each record has a *description*, a *sample name*, *keywords* and the *transcription* of the clinical note. The transcription holds most of the text (median 387 words, against 4–19 for the other fields) but decides the least: identical transcriptions carry different specialties, while the first keyword names the specialty and the section headers of the transcription reveal the type of document (Section 2.1). Hence our research question (RQ):

RQ: *Does a short, information-dense input (keywords, sample name, section headers) classify clinical records better than the complete record, and does it still do so when the keywords are missing?*

Short inputs avoid dilution: in a term frequency–inverse document frequency (TF-IDF) vector of unit length, the first keyword weighs little among the hundreds of words of a complete record. But a model that relies on the keywords learns a shortcut, which fails on the 19.6% of test records without keywords. We therefore compare short and complete inputs, on records with and without keywords.

[1] `https://www.kaggle.com`

2 Data

2.1 Data Analysis

The classes are imbalanced (Table A1): Surgery has 890 records (33.4%) and Dermatology 25 (0.9%), so always predicting Surgery gives 33.4% accuracy but F1 = 0 for the other 11 classes.

O1: unclosed quotes merge records. Seven truncated lines hold only a label, an opening quote and the start of a description, so a CSV reader merges everything up to the next quote, up to 12 records, into one field; three merged texts are cut at 32,759 characters (just under Excel's cell limit), and their rest is read as a record whose label is text. Hence `pandas` returns 2,616 rows and 15 "labels". Parsing the file line by line, we drop the seven fragments and rejoin the cut texts, recovering 2,662 records.

O2: the first keyword is the label in all 2,219 training records with keywords ("*neurosurgery*, herniated nucleus pulposus, . . . "); 80% of the test records also start with a specialty name, and the others have no keywords. Missing keywords depend on the class (Table A1): 66% of Psychiatry-Psychology records against 4% of Ophthalmology.

O3: the same transcription under several labels. 58% of the training records share their transcription with another record: 733 shared transcriptions, 732 of them under different labels (Figure A1). The labels follow two criteria: some name the type of document (84% of Surgery

records are procedure notes, with headers such as “ANESTHESIA:”; 80% of Radiology records are imaging reports, with “IMPRESSION:”), others a body system (Orthopedic, Ophthalmology), so one operation report fits both.

2.2 Data Augmentation

After an 80/20 split, the training part has 712 Surgery records but only 20 Dermatology records. We add new versions of random records of each other class until every class has 712; validation and test records are never changed.

The new versions are made by *clinical EDA*, our adaptation of Easy Data Augmentation (EDA; Wei and Zou, 2019). EDA changes a text with four random word operations; we keep two and adapt them to clinical notes. (i) *Sentence dropout*: instead of single words, we delete whole sentences, each with probability $p = 0.2$, so the remaining sentences keep their meaning (*no fracture* never loses its *no*). (ii) *Abbreviation replacement*: instead of WordNet synonyms, which cover little medical vocabulary, we swap 30 frequent abbreviations with their full form, or the reverse, each occurrence with $p = 0.5$ (MRI $\leftrightarrow$ *magnetic resonance imaging*). We drop random swap and insertion, as they create word sequences that never occur in real notes.

On the complete record (Table A2), Multinomial Naive Bayes (MNB) without balancing gets F1 = 0 for Dermatology; new rows raise its macro-F1 from 0.18 to 0.53. Exact copies score like clinical EDA, because a new row stays 93% similar to its original (cosine similarity of their TF-IDF vectors). We use clinical EDA for MNB, and class weights for the linear models, which balance the classes without new rows: the error on each record is multiplied by $n/(12\,n_c)$, where n_c is the size of its class and n that of the training part. Class weights score like clinical EDA for Logistic Regression (LR) and for the Support Vector Machine (SVM) on the keywords, but not for the SVM on the complete record (0.52 against 0.62).

3 Models

All models use TF-IDF of word 1-2-grams (lowercase, terms in at least two records, sublinear term frequency $1 + \log tf$, at most 50,000 terms) and three classifiers: MNB (the shared-task baseline family), LR and a linear SVM. MNB uses clinical EDA, and LR and the SVM use class weights (Section 2.2). Word 1-2-grams gave a higher accuracy, macro-F1 and lowest per-class F1 than word 1-3-grams and character 3-5- and 3-6-grams. The only pre-processing is lowercasing: stemming and lemmatisation yielded poorer results, probably because the keywords and the section headers work better as whole words. Each record becomes one text of “field: value” pairs. We also extract the section headers of the transcription (capital words ending in a colon, such as “PROCEDURE:”) and add them as tokens, as a short representation of the transcription. We compared several inputs: the keywords only; the complete record, directly against the complete record without the transcription (LR and SVM); and these two again without the keywords. The results (Sections 5 and 6) led to our final model, which combines two models: if a record has keywords, an SVM trained on the keywords alone predicts it; otherwise, an LR trained on the description, sample name, transcription and headers does.

4 Experimental Setup

We use the 2,662 training records of Section 2.1, and the 398 test records only to write `results.txt`. The training records are split in a stratified way: 80% for training (2,129) and 20% for validation (533), seed 42; the validation part is our own test set. A random split mirrors the test set, where 60% of the transcriptions also occur in training (50% in our validation set, O3). The submitted models are retrained on all 2,662 records. As the classes are imbalanced, we report the macro-F1 and the lowest per-class F1 (the automatic evaluation requires all

Table 1: Validation scores (split 42): accuracy (Acc), macro-F1 (F1), lowest per-class F1 (Min). *Without keywords*: field removed.

	SVM			LR		
Input	Acc	F1	Min	Acc	F1	Min
With keywords						
Keywords only	.841	.823	.167	.782	.764	.167
Complete record	.482	.516	.154	.533	.571	.364
Without transcription	.792	.764	.636	.666	.667	.372
Without keywords						
Without transcription	.356	.404	.057	.448	.492	.238
Complete record	.371	.424	.105	.473	.528	.315

above 25%) next to the accuracy, and score the final model separately on the 446 validation records with keywords and the 87 without.

4.1 Parameters/Hyperparameters

LR: $C = 1$ (inverse strength of the regularisation) and at most 2,000 solver iterations; SVM: $C = 0.5$ (cost of training errors); MNB: $\alpha = 1$ (add-one smoothing); balancing as in Section 2.2. Library: scikit-learn 1.9.1 (`scikit-learn.org`).

5 Results

Table 1 shows the validation scores of each input. The final model reaches an accuracy of 0.90 and a macro-F1 of 0.88, with an F1 per class from 0.76 (General Medicine) to 0.97 (Surgery; Table A1); Figure A2 shows its confusion matrix.

6 Discussion

As Table 1 shows, with keywords the transcription makes the results worse (SVM: 0.516 macro-F1 with it, 0.764 without), but without keywords it helps (LR: 0.528 with it, 0.492 without). With keywords, a large addition such as the transcription dilutes the impact of the keywords, and it is shared by records of different classes (O3); without keywords, it is still beneficial, because the other fields are not descriptive enough. This answers the RQ: a short input is better with keywords, but not without them. It also explains the final model: the SVM is best on the keywords (0.823 against 0.764) and LR without them (0.528 against 0.424).

A keywords-only model fails without keywords: the SVM gives all 87 validation records without them Psychiatry-Psychology, the class most often without keywords (O2), with an accuracy of 0.08.

Of the 51 errors of the final model (Figure A2), only 5 are among the 446 records with keywords: the first keyword is the gold label (e.g., “surgery, ruptured globe”), but the other keywords outweigh it; a lookup of the first keyword would avoid them. The other 46 are among the 87 without keywords, 23 of them predicted as General Medicine (precision 0.63). Most follow from O3 (record: excerpt, gold $\rightarrow$ predicted): (i) 2660: “Right shoulder arthroscopy”, Surgery $\rightarrow$ Orthopedic; the same operations appear under both labels (Figure A1), so the prediction is defensible. (ii) 2088: “Laparoscopic left adrenalectomy”, Cardiovascular-Pulmonary $\rightarrow$ Gastroenterology; probable label noise, and the prediction is sensible. (iii) 1503 and 766: “Falls at home. Anxiety and depression”, Psychiatry-Psychology and Neurology $\rightarrow$ General Medicine; the records are identical except for the label, so any classifier misses one. **Limitations:** one validation split of 533 records (5 Dermatology, 8 Psychiatry-Psychology) may give optimistic scores, and noisy or conflicting labels cap the accuracy of any classifier.

7 Future Work

For the records without keywords (46 of the 51 errors), we would try feature-based transfer learning with a pre-trained model such as BERT, reading the section headers instead of the transcription, which exceeds BERT’s limit of 512 tokens in at least 34% of the records (our initial plan).

Bibliography

Use of LLMs. We used Claude (Anthropic) in three ways: (i) to suggest how to analyse the dataset, e.g., counting the ";" separators of each line to find the broken records; (ii) to suggest data augmentation strategies, which led to clinical EDA; (iii) to draft and polish the text of the report from our notes and results, making it clearer and improving the English.

Wei, J. and Zou, K. (2019). EDA: Easy data augmentation techniques for boosting performance on text classification tasks. In *Proceedings of EMNLP-IJCNLP 2019*, pages 6382–6388.

Appendix A: Figures and Tables

Table A1: Per class: training records (n), % without keywords, % with a shared transcription, median words per transcription, main document type (P: procedure, I: imaging/test, C: consultation) and validation F1 of the final model.

Class	n	No kw	Shared	Words	Type	F1
Surgery	890	7	61	405	P 84	.97
Cardio-Pulm	308	25	58	358	P 37	.89
Orthopedic	300	16	71	505	P 60	.93
Radiology	222	9	73	230	I 80	.93
Gen. Medicine	211	42	8	420	C 74	.76
Gastro	195	15	60	285	P 54	.86
Neurology	184	27	55	419	I 48	.79
Obs-Gyn	135	17	65	387	P 52	.90
Neurosurgery	78	15	92	538	P 87	.91
Ophthalmology	73	4	58	323	P 68	.94
Psych	41	66	20	661	C 66	.82
Dermatology	25	20	48	254	P 44	.89
All	2,662	17	58	387	P 52	.88

Table A2: Balancing (validation, split 42): macro-F1 (F1) and lowest per-class F1 (Min). MNB has no class weights.

	MNB		LR		SVM	
Balancing	F1	Min	F1	Min	F1	Min
Complete record						
None	.183	.000	–	–	–	–
Class weights	–	–	.571	.364	.516	.154
Exact copies	.527	.312	–	–	–	–
Clinical EDA	.532	.293	.581	.256	.620	.211
Keywords only						
None	.296	.000	–	–	–	–
Class weights	–	–	.764	.167	.823	.167
Clinical EDA	.456	.165	.778	.167	.822	.167

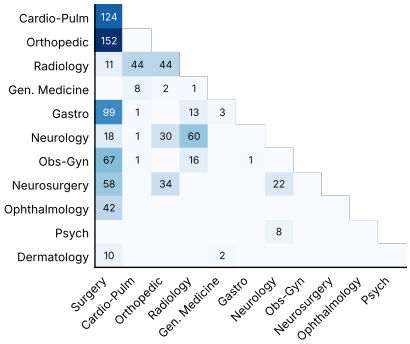


Figure A1: Identical transcriptions with different labels, per pair of classes (training set, O3).

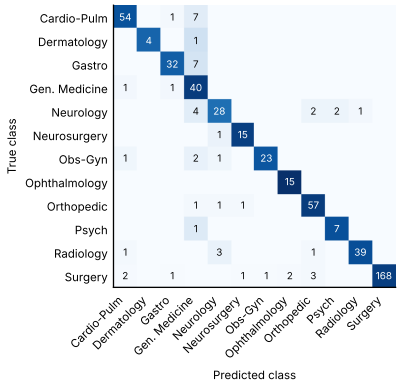


Figure A2: Confusion matrix of the final model (validation, split 42). Numbers: records; colour: share of the true class.